

The T7 proposition assembled (v1.1): $F[Q] > F[G_*]$ for all physical

TECT verification-first repository · claim B2-PROPA-HLAYER

first issued 2026-06-10 · this version issued 2026-06-10 · v1.1

1. Purpose and scope: The T7 target

Proposition (T7 target). Let $\mathcal{C}_{\text{phys}}$ be the full physically admissible competitor class (real antipodal condensate patterns at the operating endpoint $I = 2 \times 10^{-3}$, $\#Q \leq n_{\text{pack}}$, pairwise $\angle \geq \theta_{\text{min}}$). Then

$$F[Q] - F[G_*] > 0 \quad \forall Q \in \mathcal{C}_{\text{phys}} \setminus \{G_*\},$$

the isotropic Gaussian–Hartree layer G_* is the strict comparison infimum over $\mathcal{C}_{\text{phys}}$, with NO external H-LAYER hypothesis.

2. Assembly (non-circular)

Blocker A – off-shell domination, NON-CIRCULAR (T-026 v1.1). The off-shell exclusion must not assume the registered class it produces. It does not: EVEN BEFORE reducing to the registered class, a resolved off-shell mode obeys, with the ADVERSARIAL $R_{\text{lead}}(20) = 0.974$,

$$\rho_{\text{off}}^{\text{ext}} \leq R_{\text{lead}}(20) \frac{r_R}{r_R + c\delta^2} = 0.974 \times 0.587 = 0.572 < 1.$$

So off-shell modes are strictly more stable than the (already stable, < 1) on-shell modes regardless of the competitor’s T' . *After* off-shell exclusion the survivors return to the on-shell registered class, where $\rho_{\text{off}} = 0.381$ and $R_{\text{lead}}(13) = 0.650 < 1$.

Blocker C – on-shell reduction (T-026 v1.1). An on-shell, packing-bounded, coherence-resolved competitor has $T'(Q) \leq 13$: any $T' > 13$ requires off-shell (excluded by A) or super- n_{pack} (excluded by packing).

Blocker B – certified thresholds (T-027 v1.1). On the $T' \leq 13$ class the selection margins are theorem-grade: diagonal isotropy strict minimum (T-016); off-diagonal stability $R_{\text{lead}}(13) = 0.650 < 1$ (margin 0.350); selection joint(13) = $\times 1.097 > 1$ (margin 0.097), with $R_{\text{max}} \leq 0.391 < 0.634$ INTERVAL-ENCLOSED and the convention DERIVED from the $(2\pi)^3$ measure (v1.1), the sunset $S = 1.13$ rigorous, the anchoring $u_{\text{eff}} > 0$ monotonicity-certified.

Combining: off-shell excluded non-circularly $\Rightarrow$ on-shell $T' \leq 13 \Rightarrow$ both margins strictly positive and theorem-grade. Hence $F[Q] - F[G_*] > 0$ for every $Q \in \mathcal{C}_{\text{phys}}$.

3. Status: T7-CANDIDATE; the residual is the operator sign-off

T7-CANDIDATE. All three blockers are theorem-grade (Blocker B closed in T-027 v1.1: convention derived, R_{max} interval-enclosed, anchoring monotone). The residual reduces to: (i) an OPTIONAL CAS interval tightening of the already-enclosed R_{max} (the full-spacing enclosure is conservative); and (ii) the OPERATOR sign-off for the B1/B2 $T6 \rightarrow T7$ flip (no-auto-T7 rule, GOVERNANCE.md). No tier flip is enacted here.

4. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*The off-shell exclusion is circular – it uses R_{lead} of the class it produces.*” ADDRESSED (v1.1): the non-circular form uses the ADVERSARIAL $R_{\text{lead}}(20) = 0.974$ valid for ANY antipodal competitor, giving $\rho_{\text{off}}^{\text{ext}} = 0.572 < 1$ before any reduction. The registered-class $\rho_{\text{off}} = 0.381$ is the post-reduction refinement, not the basis.
- (β) “*The selection margin 0.097 is thin.*” ACKNOWLEDGED: on the on-shell $T' \leq 13$ class the joint is $\times 1.097$ (sunset-limited), comfortable relative to the excluded off-shell razor-thin case; it is the genuine operating margin.
- (γ) “*Is this enacted T7?*” HONESTLY NO: it is a T7-CANDIDATE. The operator sign-off remains. B1/B2 stay T6.

Quantitative sanity check (mandatory): A (non-circular) $\rho_{\text{off}}^{\text{ext}} = 0.572 < 1$, $\rho_{\text{off}} = 0.381 < 1$; C $T' \leq 13$; B $R_{\text{lead}}(13) = 0.650 < 1$, $\text{joint}(13) = \times 1.097 > 1$, $R_{\text{max}} \leq 0.391 < 0.634$; *margins* 0.350, 0.097 > 0.

5. Result footer

Result ID: B2-PROPA-HLAYER / T7 proposition assembly
(v1.1, non-circular)

Precise statement: T7-target: $F[Q] > F[G_*]$ for all Q in C_{phys} ,
no external H-LAYER
hypothesis. Blocker A NON-CIRCULAR:
 $\rho_{\text{off}}^{\text{ext}} \leq R_{\text{lead}}(20)$
 $r_R/(r_R + c \delta^2) = 0.572 < 1$ (adversarial,
before reduction) $\Rightarrow$
off-shell excluded; survivors on-shell (C) with
 $T' \leq 13$; on that
class (B, T-027 v1.1) $R_{\text{lead}}(13) = 0.650 < 1$,
 $\text{joint}(13) = \times 1.097 > 1$,
 R_{max} interval-enclosed < 0.634 (convention
derived), sunset rigorous,
anchoring monotone, diagonal isotropy strict
min. Margins 0.350,
0.097 > 0.

Scope: Operating endpoint $I = 2e-3$; ROBUSTNESS-MU2
band. T7-CANDIDATE.

Dependencies: offshell-domination-theorem v1.1 (A,C);
blocker-b-hardening v1.1
(B); isotropy-infimum-core v1.1 (T-016);
classwide-secondcumulant-
stability (T-020); scscope-classwide-endpoint
(T-021).

Evidence grade: ASSEMBLY (theorem-grade blockers) + EXECUTED
(res5_028_t7_proposition_assembly.py v1.1 5/5).

Reproduction command: python
codes/vacuum/res5_028_t7_proposition_assembly.py

Expected output: $\rho_{\text{off}}^{\text{ext}} = 0.572 < 1$; $\rho_{\text{off}} = 0.381$; $T' \leq 13$;
 $R_{\text{lead}}(13) = 0.650 < 1$;
 $\text{joint}(13) = \times 1.097 > 1$; margins 0.350/0.097; 5/5
PASS.

Falsification gate: A physical competitor in C_{phys} with
 $F[Q] \leq F[G_*]$; or $\rho_{\text{off}}^{\text{ext}} \geq 1$
(would need $R_{\text{lead}}(20) * r_R/(r_R + c \delta^2) \geq 1$);

or a re-opened blocker.

Tier before / after: No tier flip. B1/B2 T6 on {H-LAYER}, B5 T5.
 The T7 target is
 ASSEMBLED as a candidate (non-circular); the
 T6->T7 flip awaits the
 operator sign-off (no-auto-T7).

No-overclaim statement: Does NOT enact T7. It assembles the three
 theorem-grade blockers
 into the T7-target with a non-circular
 off-shell exclusion; the
 residual is the operator sign-off (+ optional
 CAS R_max tightening).
 T7-CANDIDATE. B1/B2 stay T6 on {H-LAYER}.

Next required action: OPERATOR sign-off for the B1/B2 T6->T7
 re-tier, closing GAP-1 (vacuum
 uniqueness) and lifting the Sectors C-F tier
 cap; optionally a CAS
 interval-arithmetic R_max bound.